INDIVIDUAL PROJECT REPORT

Review 1 — Problem Identification & Proposal (24/8/2026)

AI-Driven Edge–Cloud Network for Cyclone Prediction and Resilient Ocean Telemetry

Student	Register No.	Team No.	My role
Sreehari	24BDS0417	_____	Sensing layer (sensor and satellite simulators, hardware nodes) and recovery of lost data

Team: Laksh Ahuja (24BDS0212), Anvay Sharma (24BCE0824), Sreehari (24BDS0417). **Repository:** https://github.com/laksh-ahuja06/AI-DRIVEN-EDGE-CLOUD-NETWORK

Summary

A cyclone-monitoring network is only as good as the data its sensors deliver. In this report I describe the problem from the sensing side: realistic sensor readings, the lossy links they travel over, and how lost readings can be recovered. I present the evidence, the 15 articles I studied, the proposed solution and my plan up to Review 2.

1. Problem Statement

Ocean buoys and coastal stations lose packets in bursts when storms bring rain fade and interference, so the monitoring centre receives gaps in the very readings that show a cyclone intensifying.

- **Why it matters:** the pressure trend is a key sign of intensification; a gap hides it.
- **Who is affected:** forecasters who interpret the data, and coastal communities whose warnings depend on it.
- **If unsolved:** dashboards show missing or stale values during the storm, and models that need continuous data cannot run. At one reading per second, a 10-minute outage leaves **600 readings** missing from a single buoy.

Objectives of My Part

Objective	Measurable success criterion
Realistic sensor readings with a storm that arrives and passes	Wind > 60 km/h and pressure < 990 hPa only during the storm
Packet loss that matches real marine links	Configurable loss rate; bursts of consecutive losses
Record the true value of every reading, even lost ones	Ground-truth file for every test run
Recover lost readings better than simple methods	Lower error than holding the last value
Low-cost hardware node sending the same message format	≤ ₹1,000 per node

2. Evidence the Problem is Real

Table 1: Evidence that the problem is real

Source	Published fact / figure	Why it matters for my part
Parri et al., 2021 [1]	A real offshore LoRaWAN link from a buoy to coastal gateways 8.33 km away was monitored for 70 days, and its signal quality was compared with the weather.	Marine links work but their quality changes with conditions, so loss must be expected.
Mekki et al., 2019 [2]	Low-power wide-area radios trade data rate for range; LoRa runs at only about 0.3–50 kbps.	Sensor messages must be small, and lost ones cannot simply be re-sent.
Engineering estimate (our calculation)	At 1 reading/s and 10 % loss, about 360 readings per hour are lost per buoy; in bursts, several consecutive readings vanish together.	The receiver must recover runs of missing values, not just single ones.

3. Literature Survey

Table 2: Literature survey (15 articles, 2018–2023)

#	Author / Year	Method used	Result	Limitation
1	Parri et al., 2021 [1]	Offshore LoRaWAN field trial	8.33 km buoy-to-shore link	Single site; no data recovery
2	Mekki et al., 2019 [2]	Comparison of LPWAN technologies	Range vs data-rate trade-offs	No handling of lost data
3	Haxhibeqiri et al., 2018 [3]	LoRaWAN survey	Capacity and duty-cycle limits	Not marine-specific
4	Jouhari et al., 2019 [4]	Underwater sensor network survey	Acoustic, RF and optical links compared	Underwater focus
5	Xu et al., 2019 [5]	Review of marine IoT	Ocean sensing platforms and	Data loss not addressed

#	Author / Year	Method used	Result	Limitation
		monitoring	links	
6	Centenaro et al., 2021 [6]	Satellite IoT survey	Coverage for remote buoys	Very low throughput
7	Esposito et al., 2022 [7]	IoT early-warning review	Common sensing architectures	Assumes data arrives
8	Babiuch et al., 2019 [8]	ESP32 for measurement and processing	Low-cost Wi-Fi sensing board	Lab study, no field test
9	Wimmers et al., 2019 [9]	Deep learning on microwave imagery	Shows what satellite channels reveal	Needs real archives
10	Che et al., 2018 [10]	GRU-D with missing-value decay	Better results despite gaps	Offline clinical data
11	Cao et al., 2018 [11]	Bidirectional RNN imputation	Strong gap filling	Needs whole series
12	Yoon et al., 2018 [12]	GAN-based imputation (GAIN)	Imputes without full data	Hard to train
13	Tashiro et al., 2021 [13]	Diffusion-model imputation (CSDI)	Probabilistic estimates	Heavy compute
14	Du et al., 2023 [14]	Self-attention imputation (SAITS)	12–38 % lower error than BRITS	Offline batches; GPU
15	Lim & Zohren, 2021 [15]	Deep time-series forecasting survey	Forecasting model families	No real-time telemetry

Gap: imputation methods (GRU-D, BRITS, GAIN, CSDI, SAITS) work offline on stored datasets with heavy frameworks; marine IoT and LPWAN studies describe links but not recovery; and sensor systems are rarely tested against realistic burst loss with the true values kept for comparison. My part will generate realistic readings and loss, record the truth so recovery can be measured, recover lost readings online, and design low-cost nodes that send the same data.

Foundational methods we also build on (not counted in the 15): the Gilbert–Elliott burst-loss channel [16], [17].

4. Proposed Solution Form

Form: coding and modelling first, low-cost fabrication later. Simulated sensor nodes and a simulated satellite station give us unlimited, repeatable storm scenarios with known ground truth, which is essential for measuring recovery. Once the software works, ESP32 nodes with a pressure sensor will replace the simulator using the same message format. A product design was rejected because the scope is a proof of concept, not a manufactured device.

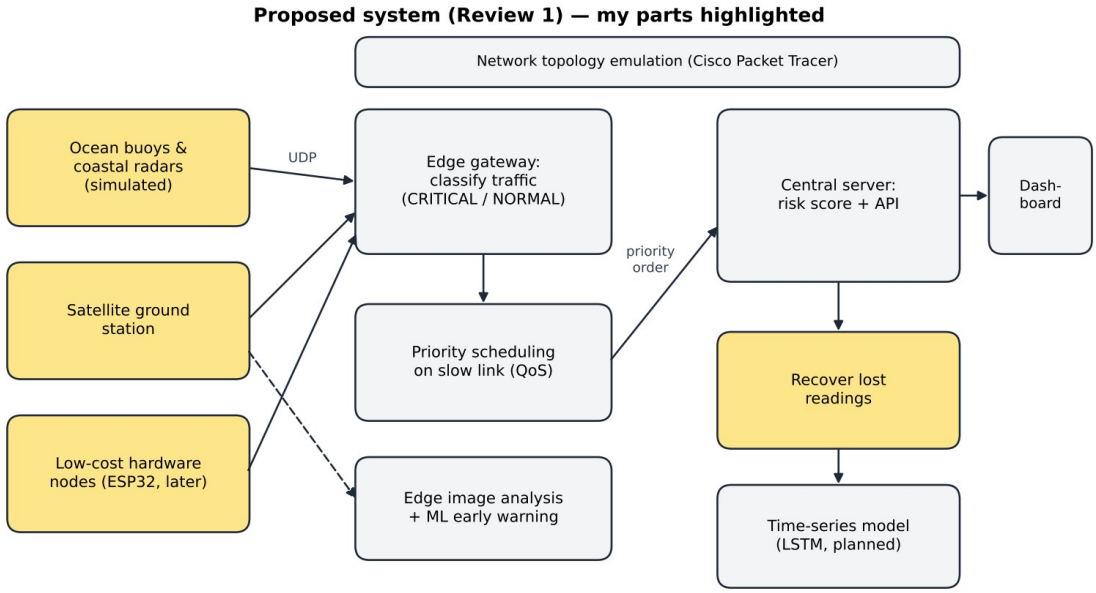


Figure 1: Proposed block diagram; my parts (sensors, satellite station, hardware nodes, recovery of lost readings) are highlighted

Planned Method for My Part

Component	Planned method
Sensor simulator	One reading per second with a sequence number, compact JSON over UDP, and a configurable loss probability.
Storm and loss models	Begin with a ramp, then a physical wind/pressure profile; begin with random loss, then a burst-loss channel.
Recovery of lost readings	Linear interpolation as the baseline, then a learned time-series model evaluated against recorded ground truth.
Hardware	ESP32 with a BMP280 pressure/temperature sensor sending the same JSON as the simulator.

5. Initial Technical Feasibility

Table 3: Feasibility of my part

Area	Status
Skills I have	Python, basic electronics, Arduino programming, statistics
Skills I must learn	Burst-loss channel modelling, time-series reconstruction, ESP32 Wi-Fi/UDP programming, I²C sensors
Tools needed / access	Python (free); Arduino IDE (free); ESP32 + BMP280 sensor (about ₹500 per node, college lab may lend); laptop
Risks	No real buoy data (mitigation: physics-inspired simulation); hardware parts delayed (mitigation: software first); recovery may not beat simple interpolation

Table 4: Risk register for my part

Risk	Likelihood	Impact	Mitigation
Hardware parts delayed or faulty	Medium	Medium	Software simulator first; borrow from lab
Simulated data too unrealistic	Medium	High	Use published storm profiles and burst-loss models
Recovery no better than interpolation	Medium	Medium	Report honestly; keep interpolation as fallback
Pressure reads wrong indoors (altitude)	High	Medium	Convert station pressure to sea level

Initial cost estimate for hardware (R1): about ₹2,000–2,500 for two nodes including breadboards and

cables; software costs nothing.

6. Team Role Split and Timeline to Review 2

Table 5: Team role split and interfaces

Team member	Area	What I hand over / receive
Sreehari (24BDS0417) — me	Sensors, satellite station, ESP32 nodes, recovery of lost readings	I define the reading format and the loss behaviour, and supply the gap-filling method
Anvay Sharma (24BCE0824)	Gateway and edge AI	Receives my packets first; needs my thresholds-crossing scenarios to test his classifier
Laksh Ahuja (24BDS0212)	Server, dashboard, QoS	Calls my reconstruction code when his server sees a missing sequence number

My role is the beginning of the data path and its repair: what the sensors send, how it gets lost, and how we recover it.

Table 6: My weekly plan from Review 1 to Review 2

Week	Dates	My milestone	Deliverable / check
1	25–31 Aug	Sensor-node simulator: readings, sequence numbers, configurable loss	Node sends JSON over UDP; LOST lines printed
2	1–7 Sep	Cyclone scenario (wind rises, pressure falls) and gap-filling baseline on the server	Missing sequence numbers are filled and marked
3	8–14 Sep	Integration test with gateway and server; hardware parts list	Prototype shown at Review 2; component list ready

R1 Video Plan

Segment	What I show
Problem and evidence (1.5 min)	Face + screen; the offshore LoRaWAN measurements and burst-loss evidence
Literature gap (1.5 min)	Face + screen; imputation and LPWAN papers; why offline imputation is not enough
Proposed solution and feasibility (2 min)	Face + screen; the sensing and recovery blocks; the ESP32 cost estimate
My role and timeline (1 min)	Face + screen; my sensor-to-hardware timeline

References

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